

Chapter 00

CTT102 – Introduction to Database



KHOA CÔNG NGHỆ THÔNG TIN
TRƯỜNG ĐẠI HỌC KHOA HỌC TỰ NHIÊN

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Chapter 0 - Introduction

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DÔNG THỊ BÍCH THUY
NGUYỄN TRẦN MINH THỦY
PHẠM THỊ BÍCH HUY
cơ sở dữ liệu

Fundamentals of
**DATABASE
SYSTEMS**
FOURTH EDITION

**DATABASE
SYSTEMS**
THE
COMPLETE
BOOK
SECOND EDITION

Hector Garcia-Molina
Jeffrey D. Ullman
Jennifer Widom

Content

- Objectives
- Learning outcomes
- Course content
- Evaluation forms and grading scale
- Reference
- Course rules

Objectives

■ Knowledge:

- Explain the roles of the database in an organization, basic concepts of database and database systems
- Data modeling: applying the ER model and relational data model to model data at a basic level
- Understanding principles of query language of database
- Understanding how to detect, describe, and declare constraints on data
- Interpret and evaluate the quality of a database schema

■ Skills:

- Design a simple database schema based on requirements
- Implement a database, use SQL to create and exploit a relational database
- Practice critical, creative thinking. Use presentation skills, English skills to read technical documents, practice teamwork skills.

Learning Outcomes

1. DB concepts: understanding the fundamental concepts of DB and DB system
2. Data modelling:
 - a. Understand concepts of Entity Relationship (ER) model - build conceptual data schema from business cases
 - b. Understand concepts of Relational data model – build a relational data schema
 - C. Transfer an ER schema to **relational data schema**
3. Query languages: proficient in relational data query languages: Relational Algebra, Relational Calculus, SQL
4. DB implementation:
 - a. Using DBMS MS SQL server to deploy a relational database schema and working with data using SQL language.
 - b. Detect, declare and implement integrity constraints in a relational database schema

Learning Outcomes

5. Quality control: assess the quality of a relational database schema and normalize the schema
6. Evolution or future directions of database systems

Content

- Chapter 1- Overview of database systems
- Chapter 2- Entity Relationship Model
- Chapter 3- Relational Data Model
- Chapter 4- Relational Algebra
- Chapter 5- SQL
- Chapter 6- Relational Calculus
- Chapter 7- Integrity Constraint
- Chapter 8- Functional Dependencies and Normalization

Assessments & Scales

- **Works in class** **25%**
 - Design ER, relational schema
 - Query languages
 - Integrity constraints
 - Functional dependencies and normalization
- **Projects/practices/lab** **30%**
 - Project or online examination
- **Final exam** **45%**
 - Multi-choice testing: 50 - 90 questions
 - Writing : 3 – 5 questions
 - Time: ~ 90 minutes
- **Extra – bonus** **10%**

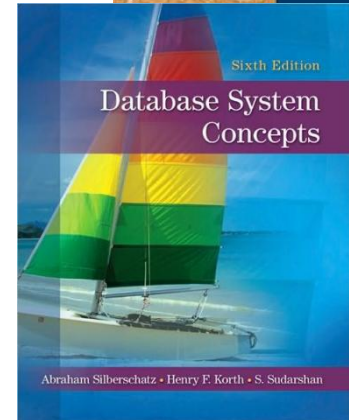
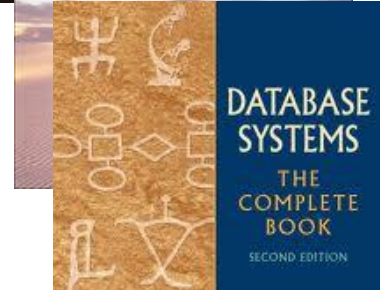
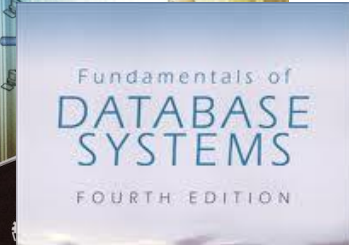
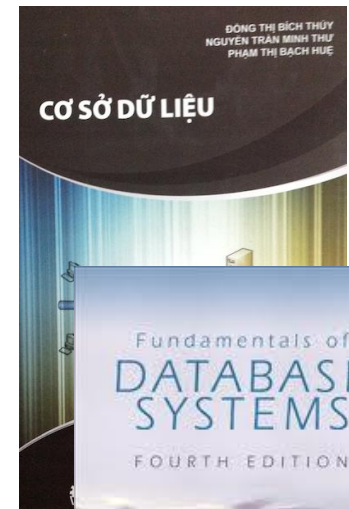
Reference

■ Vietnamese:

- **Giáo trình Cở sở dữ liệu**, Đồng Thị Bích Thủy, Phạm Thị Bạch Huệ, Nguyễn Trần Minh Thư, Nhà xuất bản Khoa học kỹ thuật, 2010.

■ English

- **Fundamentals of Database Systems**, Ramez Elmasri, Shamkant B. Navathe, Addison Wesley, 7th Edition, 2016.
- **Database Systems: The Complete Book**, Hector Garcia-Molina, Jeffrey D. Ullman, Jennifer Widom, Prentice Hall, 2000.
- **Database system concepts**, Abraham Silberschatz, Henry F. Korth, S. Sudarshan, McGraw-Hill, 2002.



Learning resource

- Website (Moodle)
 - Notifications, exchange and discussion forum, slides, exercises, assignments, etc.
- Learning resource
 - Slides
 - Theoretical exercises
 - Practical document guides
 - Reference
- Moodle link: <https://courses.ctda.hcmus.edu.vn/>

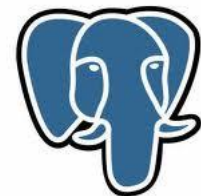
Tools and software

- MS SQL Server:

- 2005
- **2008**
- 2012
- 2016



PostgreSQL



Requirements/ Rules

■ Preparation for online sessions

- Download Zoom.us Client or Microsoft Teams and install on your PC for online courses if required
- Register using email fit, do not use other email

■ Rules

- Students are **not allowed** to be absent more than **30%** of the total class time (i.e., more than **3 sessions**).
- If two students submit identical exercises or exams, **both will receive a score of 0**.
- The final exam score must be **greater than 2** to pass.
- Do not record online lessons yourself. Any video recordings will be made and uploaded to Moodle at the teacher's discretion.
- Do not share any class videos with others.

